CS315 · ALGORITHMS ANALYSIS & DESIGN

Final Drill Practice

One drill on Lectures 1 to 5 in the midterm format

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01 Questions

Q1 MCQ · 5 PTS

1. What is the time complexity of two nested loops, each running n times?

a) O(n) b) O(n log n) c) $O(n^2)$ d) $O(2^n)$

2. The worst-case running time of merge sort is:

a) O(n) b) O(n log n) c) $O(n^2)$ d) O(log n)

3. What is the running time of the Huffman encoding algorithm?

a) O(log n) b) O(n) c) O(n log n) d) $O(n^2)$

4. What is the best-case time complexity of linear search?

a) O(1) b) O(n) c) O(log n) d) $O(n^2)$

5. Which problem needs dynamic programming because greedy does not always give the optimal answer?

a) Fractional knapsack b) 0-1 knapsack c) Activity selection d) Huffman coding

Q2 PRIMITIVE OPERATIONS · 2 PTS

Count the total primitive operations of the following algorithm in the worst case and in the best case, then give its Big-Oh.

```
Algorithm countPos(A, n)
    c ← 0
    for i ← 0 to n − 1 do
        if A[i] > 0 then
            c ← c + 1
    { increment counter i }
    return c
```

Q3 SORTING · 4 PTS

a) Apply insertion sort to the array [4, 7, 1, 8, 3, 5]. Show the array after each iteration of the outer loop.

b) Apply merge sort to the array [6, 3, 8, 1, 5, 2, 7, 4]. Show the divide steps and the merge steps.

c) For n = 8, in the recursion tree of merge sort: how many levels are there? How many problems are at level 2 and what is the size of each? What is the total work if each level costs 6n?

Q4 HASH TABLE · 5 PTS

Insert the keys 20, 34, 45, 70, 56, 13 (in this order) into a hash table of size N = 11 with h(k) = k mod 11 using:

a) Linear probing

b) Double hashing with d(k) = 7 − (k mod 7)

c) Separate chaining

0	1	2	3	4	5	6	7	8	9	10

0	1	2	3	4	5	6	7	8	9	10

Index	Chain
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Q5 LCS · 4 PTS

Using dynamic programming, find the length of the LCS of X = BCADB and Y = ABDCB, and give one LCS.

	⌀	A	B	D	C	B
⌀						
B						
C						
A						
D						
B						

Q6 MATRIX CHAIN · 4 PTS

Find the minimum number of scalar multiplications and the optimal parenthesization for A = 6×4, B = 4×10, C = 10×2, D = 2×8.

	A	B	C	D
A				
B				
C				
D				

Q7 GREEDY · 4 PTS

a) A bag holds W = 15 kg. Items (weight, value): A (5, 30), B (10, 40), C (3, 18), D (8, 24). Using the greedy method for the fractional knapsack, which items do you take and what is the total value?

b) Activities (start, finish): a1 (1, 3), a2 (2, 5), a3 (4, 6), a4 (6, 8), a5 (5, 9), a6 (8, 10). Using the greedy activity-selection algorithm, find the maximum set of compatible activities.

Q8 HUFFMAN · 2 PTS

Build the Huffman tree and the code of each character for the frequencies A = 10, B = 4, C = 7, D = 2, E = 12. How many bits does the encoded text need?